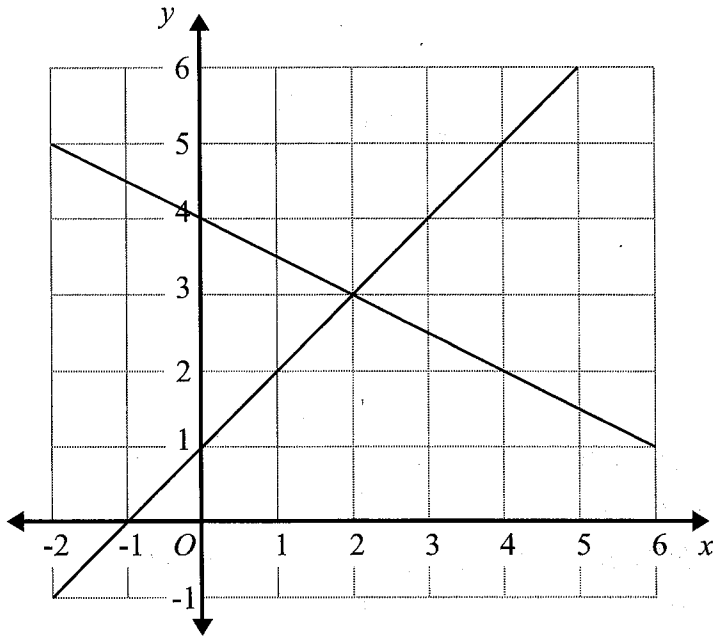
1 Write down the exact value of sin (30°)

$\frac{1}{2}$

(Total for Question 1 is 1 mark)

2 The graphs of the straight lines with equations $y = x + 1$ and $2y = 8 - x$ have been drawn on the grid.



Use the graphs to solve the simultaneous equations

$y = x + 1$
$2y = 8 - x$

$x = 2 \quad y = 3$

(Total for Question 2 is 2 marks)

3 Given that $a : b = 5:4$ and $a : c = 4:3$

Find the ratio $a : b : c$
Give your answer in its simplest form.

a : b

5 : 4

× 4

20 : 16

a : c

4 : 3

× 5

20 : 15

a : b : c

20 : 16 : 15

20 : 16 : 15

(Total for Question 3 is 2 marks)

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4 It takes 5 painters 6 days to complete a job.

Work out how many days it would take 3 painters to complete the same job.

5 × 6 = 30 days of work needed

$\frac{30}{3}$ = 10 days

10 days

(Total for Question 4 is 2 marks)

5 Solve the simultaneous equations

$$3x + 5y = -4$$
$$3x - 2y = 31$$

$7y = -35$

$y = -5$

$3x + 5(-5) = -4$

$3x - 25 = -4$

$3x = 21$

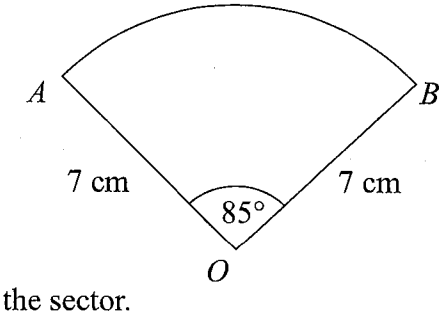
$x = 7$

$x =$ .....7.....

$y =$ .....−5.....

(Total for Question 5 is 3 marks)

6 AOB is a sector of a circle, centre O and radius 7 cm.
The angle of the sector is 85°.



Find the **perimeter** of the sector.
Give your answer correct to 3 significant figures.

Arc length $= \frac{85}{360} \times 2\pi(7)$

$= 10.4$ cm

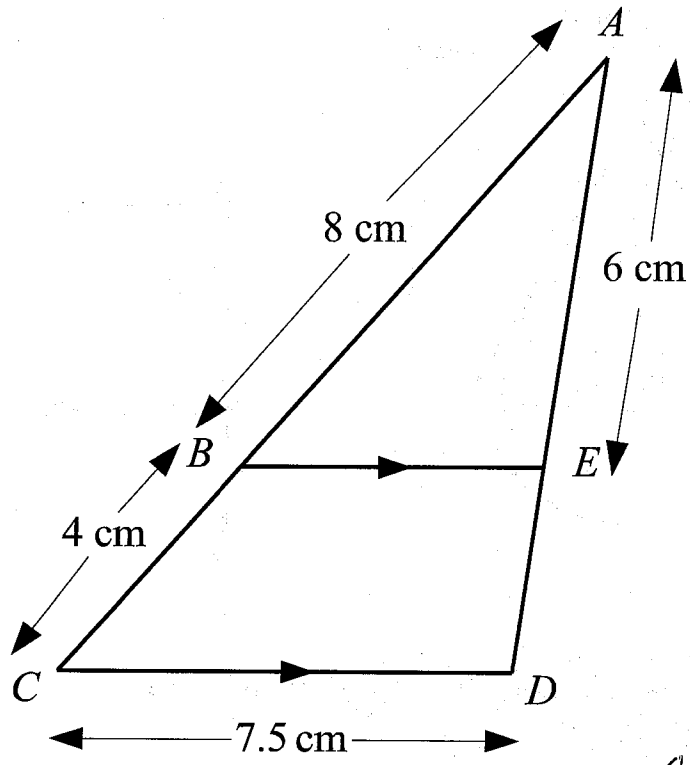
Perimeter $= 2(7) + 10.4$

$= 24.4$ cm

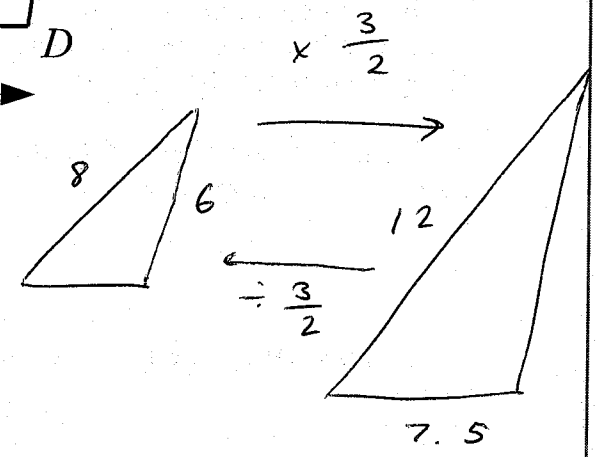
.....24.4..... cm

(Total for Question 6 is 3 marks)

7



Scale factor $= \frac{12}{8} = \frac{3}{2}$



BE is parallel to *CD*.
AB = 8 cm, *BC* = 4 cm, *CD* = 7.5 cm, *AE* = 6 cm.

(a) Calculate the length of *BE*.

$7.5 \div \frac{3}{2} = 5$

5 cm

(2)

(b) Calculate the length of *ED*.

$AD = 6 \times \frac{3}{2} = 9$

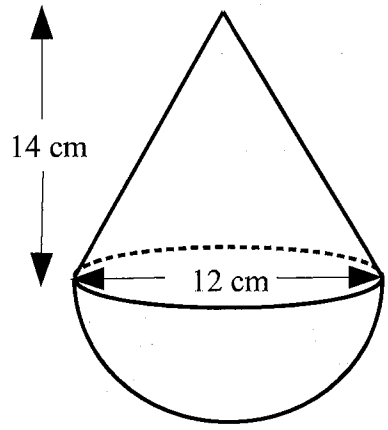
$9 - 6 = 3$ cm

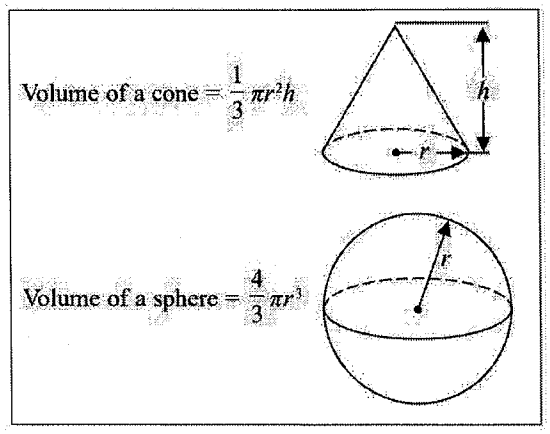
3 cm

(2)

(Total for Question 7 is 4 marks)

8 The diagram shows a solid shape.
The shape is a cone on top of a hemisphere.





The height of the cone is 14 cm.
The base of the cone has a diameter of 12 cm. $r = 6$ cm
The diameter of the hemisphere is 12 cm.

Work out the total volume of the solid shape.
Give your answer in terms of π.

volume of cone $= \frac{1}{3}\pi(6)^2(14)$

$= 168\pi$ cm^3

volume of hemisphere $= \frac{2}{3}\pi(6)^3$

$= 144\pi$

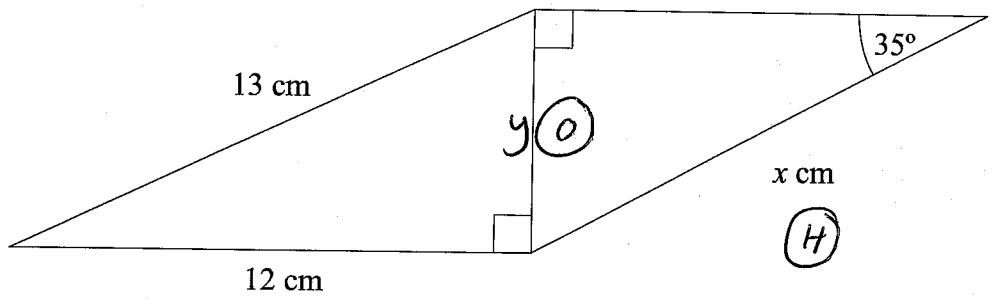
Total volume $= 168\pi + 144\pi$

$= 312\pi$

312π cm^3

(Total for Question 8 is 4 marks)

9



Work out the value of x.
Give your answer to 1 decimal place.

$12^2 + y^2 = 13^2$

$y^2 = 13^2 - 12^2$

$y^2 = 25$

$y = \sqrt{25}$

$= 5$ cm

$\sin\theta = \frac{O}{H}$

$\sin(35) = \frac{5}{x}$

$x \sin 35 = 5$

$x = \frac{5}{\sin(35)}$

$= \underline{\underline{8.7 \text{ cm}}}$ 1dp

8.7

(Total for Question 9 is 4 marks)